

IMO Training – Number Theory

Assignment 1

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- Each question carries 7 marks.
- Complete solutions with justified arguments are required for full credit in every question.
- Please scan your work and submit your work (in pdf format) through email.
- Deadline for submission: 15 August, 2020

1. (Hong Kong TST 1997)
Prove that there are infinitely many $n \in \mathbb{N}$ s.t. $n \mid 2^n + 1$. Find all such n 's that are prime numbers.
2. (Hong Kong TST 2000)
Prove that $\forall n \in \mathbb{N}, 46410 \mid n^{30} - n^{18} - n^{14} + n^2$.
3. (Hong Kong TST 2001)
Find all $n \in \mathbb{N}$ s.t. 72 does not divide $n^8 - n^2$.
4. (Hong Kong TST 2003)
Show that $\exists k \in \mathbb{N}$ s.t. $\forall n \in \mathbb{N}, k \cdot 22^n + 1$ is not a prime.
5. (Hong Kong TST 2004)
A positive integer n is said to be *good* if $\exists a, b \in \mathbb{N}$ s.t. $a + b = n$ and $n \mid ab$. For instance, 9 is good because $3 + 6 = 9$ and $3 \cdot 6$ is divisible by 9. Furthermore n is said to be *very good* if we can choose a, b to be distinct. Hence 9 is very good.
 - (a) Given that 2003 is a prime, show that 2003 is not good.
 - (b) Is 2002 good?
 - (c) Find all positive integers which are good but not very good. Justify your answers.